Sixth Monitoring Committee Report

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January 2024

1 Progress summary

I continue the work on my research project after confirming my PhD registration in June 2023. In the last term I have worked on the following tasks:

(1) Prepared and submitted a research paper to ACM SIGIR Conference on Human Information Interaction and Retrieval (CHIIR), based on the work completed to date (draft can be seen here). The paper has been rejected, but will be resubmitted to the IEEE World Congress on Computational Intelligence (WCCI) after small adjustments.

(2) Performed a literature review on LLM based Conversational Recommender Systems (CRS), which is written up in this report and will form a part of the thesis chapter.

(3) Proposed a system design that integrates LLMs with CRS components completed to date and started initial experiments with in-context learning.

Results of this work are described in detail further in this report.

2 Literature Review

2.1 Introduction

In recent years, there has been a notable rise in the popularity of chat-based assistants, particularly with the widespread adoption of Large Language Models (LLMs). Dedicated tracks for Conversational Assistants at TREC conferences (2019-2023) and the recognition of "Dialogue and Interactive Systems" as a major research topic at SIGIR highlight the research community's response to this trend.

This increased interest extends to natural language-based conversational recommender systems (CRS). The incorporation of natural language in recommender systems is not a novel concept, as evidenced by the three-decade-long utilisation of text associated with items, such as item descriptions and reviews, in content-based recommenders [1].

Large scale non-interactive and point-and click recommender systems have well known challenges such as promoting clickbait, perpetuating societal biases, and contributing to the polarisation of the user base [2]. Chat based CRS help address these challenges by providing opportunities for users to interactively explore system suggestions and supplement implicit signals with explicit intents. The use of natural language interfaces enables users to express their preferences in ordinary language, allowing for a more general-purpose description of their preferences [1]. These user requests can be embedded in the same space as recommendable items and their metadata, facilitating similarity-based retrieval using any combination of items, attributes, or contextual information, such as user goals, mood, or situation [3].

Moreover, interaction with CRS can be enriched with additional capabilities, including review summarization and explanation generation [4]. Recent works highlight the use of pre-trained generative language models (PLMs) as a versatile engine for various recommendation-related tasks, as exemplified by the utilisation of PLMs in building unified CRSs capable of performing multiple tasks with a single model, as opposed to using multiple components [5].

In recent months several research groups proposed CRS architectures based on Large Language Models (LLMs), such as ChatGPT [6], LaMDA [2] and LLaMa [7]. LLMs are transformer based language models pre-trained on diverse and extensive datasets, which allows them to have a broad understanding of language patterns, grammar, and semantics without or with only minimal fine-tuning. Furthermore, LLMs possess inherent knowledge about the world and the ability to engage in logical reasoning.

Recent studies leverage the reasoning capabilities of LLMs to supplant more conventional approaches in recommendation systems. The potency of LLMs enables the resolution of longstanding challenges in CRS, such as limited natural language understanding, absence of content and context knowledge, templated or restricted conversations, reactive approach to conversation and overall system complexity.

Natural language understanding. While traditional CRS are typically capable of recognising entities in context using an external knowledge base, challenges persist in comprehending the semantic information within the context [5]. LLM based CRS prove more adept in addressing these understanding challenges [5] and have been shown to effectively capture diverse forms of user preferences, encompassing previously enjoyed items, topics, current or desired emotions, preferred company for movie-watching, and free-form requests [8].

Built-in content and context knowledge. LLM based CRS have been shown to be beneficial in cold start settings, which is attributable to their superior content knowledge and generalisation ability [9]. While they generally exhibit weaker collaborative knowledge compared to CRS specifically trained on collaborative datasets, they surpass them in content knowledge [9]. This strength in content knowledge proves particularly valuable in attribute-centred natural conversations. Interestingly, LLM generalisation capability extends across domains. LLMs can migrate user preferences across domains, enabling them to achieve cross-domain recommendations, e.g, recommending books and activities based on movie preferences [6, 7]. Moreover, LLMs showcase an unprecedented ability to tap into general world knowledge and employ some level of common-sense reasoning [2].

User intent diversity. In natural conversations, users often make requests beyond seeking recommendations. They may expect human-like responses, such as preference refinement, knowledgeable discussion, or justification for a recommendation [9]. Users express these intents over the course of a full multi-turn conversation, and CRS must understand and connect these intents to the item corpus [2]. This introduces multiple recommendation-related subtasks, leading to challenges in managing user intents, solving individual tasks, and generating satisfactory responses. In the past year, LLMs have been shown to have a capability to successfully handle a variety of recommendation related subtasks, such as item ranking, providing information about items, justifying recommendations to users and refining recommendations based on user feedback [8, 10]. On top of that, LLMs have been used as a dialogue management component that triggers appropriate subtask components, allowing researchers to build systems capable of understanding and responding to more complex and diverse user intents [10]. Therefore, combining recommender systems with LLMs goes beyond replacing existing approaches and enhances the overall recommendation process [11]. The more natural conversations facilitated by LLMs result in higher user satisfaction and engagement [12].

Mixed initiative conversation. Traditional CRS are typically user-led, sharing a significant weakness with many conversational agents in responding only passively to user queries, which limits their ability to offer high quality recommendations based on a broader context than a given conversation [13]. These models are lacking proactiveness and are not able to initiate a conversation, shift topics, or offer recommendations considering a more extensive context [13]. LLM-based solutions, on the other hand, exhibit the capability for a mixed-initiative setup. In this setup, the system not only responds to user requests but also actively steers the conversation in a specific direction. An example is preference elicitation, where the system determines when and how to best query the user to extract information about their preferences [2].

System simplicity and flexibility. Traditional CRS typically incorporate multiple purpose built and especially trained components in order to perform recommendation related subtasks (natural language understanding and generation, dialogue management, recommendation, preference elicitation etc.). An LLM can perform various actions and stand for different components in a CRS with minimal fine-tuning or prompt engineering [4]. This allows to streamline and simplify the process of CRS development and shift the focus from algorithm engineering problems towards prompt engineering and dataset generation [2].

2.2 Overview of LLM based CRS

2.2.1 LLM Applications in CRS

2.2.1.1 Recommendation Task

Studies primarily investigate the effectiveness of LLMs when directly employed as a recommendation component whose output is a ranked list of items [3, 9]. Typically, in these works LLM is treated as a content-based recommender that takes a combination of desirable item attributes, description of likes and dislikes and user profile information, including past behaviour, as an input [1, 3, 9, 10]. However, some studies focus on item-based or sequential recommendations, where the chronologically ordered history of previous interactions with items is the main predictor [1, 7, 8]. Comparative studies show that combining language and item-based preferences does not provide a performance boost and item-based LLM methods can perform on par with classical collaborative filtering methods.

In practice, an external model is commonly employed to retrieve a shortlist of recommendations, subsequently re-ranked by an LLM [8]. While LLMs generally possess intrinsic knowledge of items in the recommendation corpus through pre-training on extensive datasets, it is likely that they have not encountered all items during training, particularly less popular or new ones. In certain scenarios, the LLM may lack familiarity with the recommendation domain altogether [2]. In the retrieval phase, the objective is to take the full corpus, which, in certain domains like videos, may contain hundreds of millions of items [2]. Given the token constraints, passing a large number of items in a prompt is not possible, therefore, the number of candidates needs to be kept low. Various retrieval and recommendation architectures, such as deep-learning based dual encoder model [2] or an an ensemble comprising a matrix factorization component and a neural component [8]have been proposed for this purpose.

2.2.1.2 Supporting Tasks

Natural conversations centred around recommendations can be complex and include a variety of interaction types between a recommendation seeker and a recommender. Users expect a CRS to perform a number of different tasks in order to enable these interactions. Some researchers even argue that the task of delivering recommendations should be executed using a visual interface, while CRS should be reserved for sustaining a natural dialogue with users and handling supporting tasks [2].

While in general LLMs offer the capability to address these expectations [11, 12], model fine-tuning and prompt engineering for each subtask individually is needed to achieve good performance of separate components and CRS as a whole [10].

To date, LLMs have been applied to various recommendation subtasks, including:

1. *Preference elicitation.* Leveraging their reasoning capabilities, LLMs can dynamically construct user profiles and generate questions in order to acquire missing information [2, 11]. LLMs have been used to query users about their preferences either directly [5] or with support of external models such as KBRD and KGSF [10].

2. *Recommendation refinement.* Some implementations utilise natural language understanding capabilities of LLMs to build critique based systems, where users can either accept recommendation or provide feedback to the CRS to refine the recommendations [8].

3. *Recommendation explanation.* Multiple authors recognise that users want to understand the rationale behind the recommendation and propose to use LLMs to generate recommendation explanations either directly [2, 8, 12] or in combination with external models, such as KBRD and KGSF methods 10.

4. *Providing item information.* Finally, LLMs are used to generate item descriptions and respond to user queries about the recommended items [8, 11].

2.2.1.3 Dialogue Management

As conversations with CRS assistants become more complex, effective dialogue management becomes a crucial challenge. Recommender centred conversations require a mixed-initiative setup, where the the system must be able to switch between responding to user requests and proactively guiding the conversation in a specific direction [2].

Due to their ability to reason, LLMs are used to handle dialogue management across diverse domains for problems such as context tracking, task planning, integrating external tools and executing system calls, such as sending requests to a recommendation engine [2, 10]. The ability of LLMs to orchestrate the dialogue flow allows LLM based CRSs to demonstrate state-of-the art results on natural recommendation datasets such as ReDIAL and INSPIRED [9].

During a system turn, an LLM generates a sequence of natural language outputs that encompasses intermediate reasoning, API calls to interact with the rest of the system, as well as user facing natural language generation [2]. Typically, markers are defined to differentiate between intermediate reasoning concealed from the user from user facing responses. For example, Friedman et al. used “Response: ” keyword to indicate that the message is intended for user presentation and “Request: ” keyword for outputs intended to be forwarded to the the recommendation engine backend, facilitating the retrieval of a slate of recommendations [2].

The sequence of LLM calls resulting in a user facing reply typically consists of multiple well defined stages [10], such as:

1. *Sub-task detection*, where LLM analyses dialogue context to identify specific sub-task to be executed in the current dialogue turn based on its knowledge.

2. *Model matching*, where LLM distributes the identified sub-task to a suitable expert model according to model descriptions provided in the prompt.

3. *Sub-task execution*, where LLM processes and summarises inference information returned by an expert model.

4. *Response generation*, where LLM acts as a language interface for response generation based on the information generated in the previous steps.

2.2.1.4 User Simulation

Finally, LLMs can be used for user simulation users in training and evaluating CRS. In CRS, interactions are dynamic and each turn of the CRS influences subsequent user behaviour, so employing a gym-like environment for evaluation is preferred over relying solely on static datasets. Friedman et all propose a number of techniques that allow to construct a controllable LLM-based user simulator, enabling the generation of synthetic conversations by mimicking diverse user behaviours and scenarios [2].

2.2.2 Prompting Approaches

The remarkable performance of LLMs on tasks on which they were not explicitly trained has led to the emergence of a novel paradigm known as In-Context Learning (ICL) [6], prompt learning [4] or prompt engineering [3]. This paradigm allows to adapt tasks to the model rather than the other way around [4].

A growing body of research endeavours to discover optimal methods of interacting with LLMs through prompting. There are three prevalent approaches: zero-shot, few-shot, and chain-of-thought [1, 2, 3].

1. *Zero-shot prompting.* The model is presented with instructions and contextual information in order to complete a task. In the domain of CRS, this could entail providing user profile information [11], natural language preference statements [1] and historical interactions [6]. The model is then expected to generate responses or recommendations based on this information without specific task-related training. For simple repetitive templates, prompt distillation, such as the PrOmpt Distillation (POD) approach, can be employed to distil the knowledge in discrete prompts into continuous prompt vectors. During training, these vectors learn the expressions in the discrete prompt, and during inference, only the continuous prompt is utilised to reduce computation cost [4].

2. *Few-shot prompting.* In addition to instructions, the model is provided with a limited number of examples showcasing a specific task as part of the input and is prompted to replicate the task by generating its own output in response to a similarly structured question. This approach, involving a small set of examples, often yields superior results compared to a one-shot learning scenario. For instance, in CRS it yields results competitive with traditional collaborative filtering methods [1].

3. *Chain-of-thought prompting (CoT).* The model is prompted to provide intermediate answers before arriving at the final solution. This approach aims to simulate a multi-step thought process akin to human reasoning when tackling complex problems. In the context of CRS, the intermediate steps in the CoT can offer explanations for the inclusion or exclusion of specific items in the list of recommendations and provide valuable insights into the model's decision-making process [2]..

2.2.3 LLM Training and Fine-tuning

2.2.3.1 Pre-trained Models

In CRS domain, commercial GPT-based models consistently outperform various open-sourced LLMs [9]. Several studies experiment with GPT-3.5-turbo and GPT-4, the models behind ChatGPT, which is typically used with no fine-tuning and can achieve good results with prompt engineering alone [3, 8, 9].

On the other hand, a wide range of open-source LLMs are used withing CRS, such as LLAMA [5, 7, 9, 10], PaLM [1], BAIZE [9], T-5 [4, 10] and LaMDA [2]. These models often require fine-tuning for optimal performance.

2.2.3.2 LLM Fine-tuning

LLMs are commonly trained using the Instruction Tuning approach. In this method, task Instructions, task inputs and task outputs are defined in natural language. Instructions and task inputs serve as model inputs and model is optimised to return desired task outputs [7]. Fine-tuning of LLMs is typically conducted using Low-Rank Adaptation (LoRA) [7]. LoRA involves freezing the pre-trained model parameters and incorporating trainable rank decomposition matrices into each layer of the Transformer architecture, enabling lightweight tuning.

Fine-tuning for downstream tasks can be carried out in a supervised fashion by adapting LLMs to task-specific datasets [4]. An alternative approach involves reinforcement learning [10]. This method, referred to as Reinforcement Learning from Performance Feedback (RLPF), utilises recommendation and conversation feedback as reward signals and employs the REINFORCE method to guide the learning direction. In the RLPF setup, the LLM is treated as the agent parameterized with Θ, and the generated solutions S are considered actions for solving conversational recommendation problems. The reward signal, R, comprises recommendation evaluation metrics such as HIT and response generation evaluation metrics like BLEU.

2.2.4 LLM Based CRS Evaluation

CRS are complex systems with multiple aspects of performance, therefore, researchers often combine different approaches and metrics for quality evaluation.

2.2.4.1 Recommendation Performance

Recommendation quality is typically assessed using metrics established in Recommender System field, such as Normalised Discounted Cumulative Gain (NDCG), Hit Ratio (HR) and Mean Reciprocal Rank (MRR) [3, 4, 8]. While these metrics have been found to correlate to qualitative user evaluation, the strength of these correlations are typically weak, e.g. a recent study reported a Pearson's coefficient of 0.26 for NDCG@9 and users' recommendation task evaluation [8]. In addition to the above metrics, some researchers consider alternative evaluation lens, such as recommendation novelty [3].

2.2.4.2 Text Generation Performance

Generative performance is evaluated in reference to a human generated dataset and uses metrics such as Bilingual Evaluation Understudy (BLEU) [10]. However, available datasets do not often exemplify high quality natural human dialogue and this form of evaluation may not be an accurate representation of the response quality.

Alternative approaches include user simulation, usually also implemented using an LLM [2] and user studies involving human participants rating satisfaction on a Likert scale [8].

2.2.4.3 Datasets

Multiple natural language based dialogue datasets are made available for CRS evaluation: INSPIRED \citep{hayati2020inspired}, GoRecDial \citep{kang2019recommendation}, TG-ReDial \citep{zhou2020towards}.

More recently, more complex and exhaustive natural datasets were proposed and used in LLM CRS training, facilitated by the increased NLU capabilities that LLMs provide. For example, He et al. constructed a Reddit-Movie dataset based on 634k naturally occurring recommendation centred dialogs from users from Reddit, a popular discussion forum [9].

2.2.5 LLM based CRS Challenges

While the integration of LLMs in CRS significantly enhances natural language understanding and the quality of generated text, several challenges persist:

1. *Item Knowledge.* Pre-trained LLMs, such as ChatGPT, cannot recommend items released beyond the timeframe of their training data. Strategies involving external information injection into prompt are explored to enable ChatGPT to handle recommendations for new items [6].In some domains, such as YouTube videos, LLM cannot possess knowledge about individual item titles and needs to interface with the corpus [2].

2. *Hallucinations.* In LLMs, "hallucination" refers to the generation of information or outputs that are not grounded in reality or the training data. Hallucinations in LLM CRS pose a challenge, as these models may occasionally generate recommendations beyond the actual content availability of the platform [2, 8]. The issue of hallucinations remains largely unsolved in LLMs, indicating the need for further research and solutions [2].

3. *Recommendation quality.* LLMs generally have weaker collaborative knowledge than traditional CRS [5, 9] and on sequential recommendation task perform similarly to random guessing [7], which can be mitigated by fine-tuning and using a stand-alone recommender component that feeds recommendation into the LLM prompt. In addition to that, LLM recommendations are susceptible to popularity bias, favoring popular items excessively [9].

4. *Preference elicitation.* LLM CRS do not typically perform well on information elicitation task, where traditional CRS demonstrate better results [5].

3 Proposed System Design

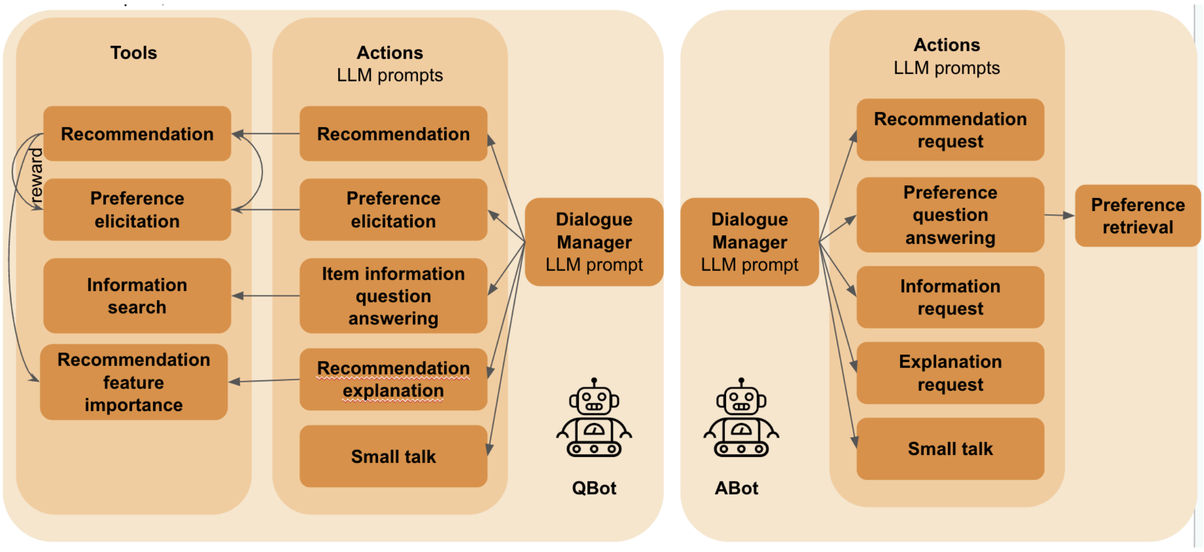

Figure 1. End-to-end natural language interface CRS

The planned system architecture is shown in Figure 1. The system will consist of pre-trained models: (1) Recommendation component and (2) Preference Elicitation component (3) LLM, that are called by LLM-based Dialogue Management module. The system will be fine-tuned through cooperative bot-play – interaction of two independent algorithms with common reward. QBot will be optimised to provide optimal recommendations from ABot, and ABot will be optimised to provide accurate responses to QBot. QBot will be blinded to ABot's preference history and will be able obtain this information only through dialogue, following which it will generate item recommendations.

3 Transferable Skills

A detailed list of courses and conferences that I have attended are listed in the Appendix 5.1 and 5.2.

4 References

[1] Sanner, S., Balog, K., Radlinski, F., Wedin, B. and Dixon, L., 2023, September. Large language models are competitive near cold-start recommenders for language-and item-based preferences. In Proceedings of the 17th ACM conference on recommender systems (pp. 890-896).

[2] Friedman, L., Ahuja, S., Allen, D., Tan, T., Sidahmed, H., Long, C., Xie, J., Schubiner, G., Patel, A., Lara, H. and Chu, B., 2023. Leveraging Large Language Models in Conversational Recommender Systems. arXiv preprint arXiv:2305.07961.

[3] Spurlock, K., 2023. Evaluating chatGPT for recommendation: how does the ability to converse impact recommendation?

[4] Li, L., Zhang, Y. and Chen, L., 2023, October. Prompt distillation for efficient llm-based recommendation. In Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (pp. 1348-1357).

[5] Liu, Y., Zhang, W.N., Chen, Y., Zhang, Y., Bai, H., Feng, F., Cui, H., Li, Y. and Che, W., 2023. Conversational Recommender System and Large Language Model Are Made for Each Other in E-commerce Pre-sales Dialogue. arXiv preprint arXiv:2310.14626.

[6] Gao, Y., Sheng, T., Xiang, Y., Xiong, Y., Wang, H. and Zhang, J., 2023. Chat-rec: Towards interactive and explainable llms-augmented recommender system. arXiv preprint arXiv:2303.14524.

[7] Bao, K., Zhang, J., Zhang, Y., Wang, W., Feng, F. and He, X., 2023. Tallrec: An effective and efficient tuning framework to align large language model with recommendation. arXiv preprint arXiv:2305.00447.

[8] Gutierrez Granada, Mateo, Dina Zilbershtein, Daan Odijk, and Francesco Barile. "VideolandGPT: A User Study on a Conversational Recommender System." arXiv e-prints (2023): arXiv-2309.

[9] He, Z., Xie, Z., Jha, R., Steck, H., Liang, D., Feng, Y., Majumder, B.P., Kallus, N. and McAuley, J., 2023, October. Large language models as zero-shot conversational recommenders. In Proceedings of the 32nd ACM international conference on information and knowledge management (pp. 720-730).

[10] Feng, Y., Liu, S., Xue, Z., Cai, Q., Hu, L., Jiang, P., Gai, K. and Sun, F., 2023. A Large Language Model Enhanced Conversational Recommender System. arXiv preprint arXiv:2308.06212.

[11] Kolb, T.E., Wagne, A., Sertkan, M. and Neidhardt, J., 2023, November. Potentials of Combining Local Knowledge and LLMs for Recommender Systems. In TBC (Vol. 3560, pp. 61-64). CEUR-WS. org.

[12] Hua, W., Li, L., Xu, S., Chen, L. and Zhang, Y., 2023, September. Tutorial on Large Language Models for Recommendation. In Proceedings of the 17th ACM Conference on Recommender Systems (pp. 1281-1283).

[13] Liao, L., Yang, G.H. and Shah, C., 2023, July. Proactive conversational agents in the post-chatgpt world. In Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (pp. 3452-3455).

5 Appendix

5.1 Transferable Skills

Type	Title	Dates	Status
Autumn 2019			
Academic conference	ACM RecSys19 - The ACM Conference on Recommender Systems	16/09/2019 - 18/09/2019	Attended
Taught module	CSMMA16 - Mathematics and Statistics	01/10/2019 - 13/01/2020	Completed, 99/100
Reading materials	Complexity Analysis	11/11/2019 - 17/11/2019	Completed
MOOC	Advanced Machine Learning Specialisation: Introduction to Deep Learning (Coursera)	16/09/2019 - 20/10/2019	Completed
Spring 2020			
MOOC	Advanced Machine Learning Specialisation: Practical Reinforcement Learning (Coursera)	11/11/2019 - 28/01/2020	Completed
Taught module	CSMAI19 - Artificial Intelligence	16/01/2020 - 11/05/2020	Completed, 95/100
Summer 2020			
Academic conference	ACL 2020 Virtual Conference	6/07/2020 - 8/07/2020	Attended
MOOC	Advanced Machine Learning Specialisation: Natural Language Processing (Coursera)	13/07/2020 - 20/09/2020	Completed
Autumn 2020			
Taught module	CSMRS16 - Research Studies	01/10/2020 - 10/01/2021	Completed, 79.3/100

5.2 RRDP Training Record

Title	Dates	Status
Autumn 2019		
How to Write a Paper	21/11/2019	Attended
Spring 2020		
An Introduction to LaTex	19/02/2020	Attended
Introduction to Bibliometrics	25/03/2020	Cancelled
Autumn 2020		
Ensuring Confirmation of Registration	25/11/2020	Attended

5.3 Training Needs Analysis

Available here.